Final Project: Track B

1 Project Overview

In this project, you will apply machine learning methods covered in this course to a biological or biomedical research question of your choice. You may use your own research data, laboratory data for which you have permission, or a public dataset.

The project should emphasize sound experimental design, model comparison, generalization, biological interpretation, and reproducibility—not merely achieving the highest prediction score. Unlike Track A, you are not required to conduct both regression and classification if your biological question supports only one task. The overall expectations and grading standards are comparable to Track A.

2 Learning Objectives

Upon completion, you should be able to:

- Formulate a meaningful biological prediction question and define predictors and outcomes.
- Assess data quality and perform exploratory data analysis and leakage-free preprocessing.
- Compare appropriate linear and nonlinear methods and justify their selection.
- Use cross-validation to tune hyperparameters and estimate development-set performance.
- Evaluate selected models on an independent held-out test set to assess generalization performance.
- Investigate overfitting, underfitting, model complexity, and regularization.
- Interpret predictive results in their biological context, with appropriate limitations.
- Communicate findings in reproducible notebooks, a scientific report, and a five-minute presentation.

3 Project Selection and Proposal Approval

Research question and task. State a specific biological or biomedical question and identify the response variable. Regression predicts a continuous outcome; classification predicts a categorical outcome. Both are allowed, but both are not required.

Dataset. Describe the source, sample size, predictor count and types, outcome distribution, biological meaning, missingness, class imbalance, and any related or repeated samples. Confirm that you have lawful access and permission to use the data. Do not upload restricted or identifiable data without authorization. The dataset must support a meaningful separation into development and independent test sets. You must describe your proposed splitting strategy in the project proposal. Projects for which a valid held-out test set cannot be established will not be approved.

Algorithms. Compare **at least three appropriate algorithms covered in the course**, chosen from linear regression, logistic regression, decision trees, random forests, and multilayer perceptrons (MLPs). Include a simple linear baseline when scientifically appropriate, with nonlinear alternatives. Explain why each method is suitable. Exceptions require instructor approval.

Analysis plan. Specify splitting and cross-validation strategy, preprocessing, primary selection metric, tuning plan, two model-complexity experiments, and biological interpretation approach. Your proposal must be feasible within three weeks. The instructor may approve, request revisions, or decline a proposal that does not meet these requirements.

4 Project Tasks

4.1 Part 1. Exploratory Data Analysis and Preprocessing (15 points)

4.1.1 Dataset exploration

Describe the number of samples and predictors, predictor types, response definition and distribution, missing values, and relevant data quality issues. Include at least two informative visualizations appropriate to the data. Discuss the biological meaning of the outcome and key predictors.

4.1.2 Data splitting

Set aside an independent test set before model development. An 80% development / 20% test split is recommended for independent samples. For classification, stratify when appropriate. If samples are grouped by patient, batch, study, family, or time, use a defensible group-aware or time-aware splitting strategy to prevent data leakage. The test set must remain untouched during preprocessing decisions, model development, hyperparameter tuning, and model selection.

4.1.3 Preprocessing

Use scikit-learn `Pipeline` objects for data-dependent preprocessing and feature selection. Explain missing-value handling, encoding, scaling, and other steps as applicable. Discuss why these steps must be learned only from training folds and how leakage is avoided.

4.2 Part 2. Machine Learning Model Development (20 points)

Implement at least three instructor-approved algorithms appropriate for the task. Include a simple baseline when applicable. Explain the assumptions and rationale for each algorithm. Fit baseline configurations, specify justified hyperparameter grids, and tune on development data using cross-validation. Use a consistent evaluation protocol across algorithms so comparisons are meaningful.

Examples of tunable parameters:

Algorithm	Candidate hyperparameters
Linear/logistic regression	Regularization strength (when applicable)
Decision tree	`max_depth`, `min_samples_leaf`
Random forest	`n_estimators`, `max_depth`
MLP	`hidden_layer_sizes`, `alpha`

A small, justified search grid is sufficient. Document random seeds, optimization settings, and any convergence warnings.

4.3 Part 3. Model Evaluation and Comparison (20 points)

Choose a primary model-selection metric *before* comparing models. For regression, consider MSE, RMSE, MAE, and R^2; RMSE is a reasonable default. For classification, consider accuracy, precision, recall, F1, and ROC AUC; choose based on the biological consequences of errors and class balance. Compute ROC AUC using continuous scores, not hard class predictions.

Report cross-validation mean and standard deviation for appropriate metrics, and independent test results for each selected algorithm configuration. After tuning, refit each selected configuration on the full development

set and evaluate on the held-out test set once. Select the preferred algorithm using development results, not test-set rankings.

The preferred model must be identified using cross-validation results before examining test-set performance. Test results should be used only for final evaluation and reporting.

Required outputs:

- A cross-validation model comparison table and a final held-out test-performance table.
- Appropriate visualizations: e.g., observed-versus-predicted and residual plots for regression; ROC curves and confusion matrices for classification.
- An interpretation of performance differences, uncertainty, error patterns, and practical relevance to the biological question.

4.4 Part 4. Model Complexity and Overfitting (10 points)

Conduct **two systematic experiments** on development data only. Suitable examples include varying decision-tree depth, MLP hidden-layer size or regularization strength, random-forest complexity, or training sample size. Choose experiments that are informative for your dataset and algorithms.

For each experiment: (1) state a specific question, (2) systematically vary a model setting while controlling other factors, (3) report training and cross-validation performance, (4) provide a clear plot, and (5) explain evidence of overfitting, underfitting, or improved generalization. Use the same evaluation metric across settings within an experiment. Do not inspect the independent test set to make these decisions.

4.5 Part 5. Model Interpretation and Biological Discussion (15 points)

Compare algorithms in terms of predictive performance, complexity, interpretability, and suitability for the research question. Examine model coefficients where applicable and use an appropriate model-agnostic method such as permutation importance on development data. Identify and discuss biologically meaningful predictors, referring to dataset documentation or scientific literature as appropriate.

Discuss relevant limitations: sample size, high dimensionality, correlated predictors, confounding, dependence among samples, potential distribution shift, and generalizability. Distinguish predictive association from causation. Explain what additional data or validation would strengthen your conclusions.

5 Required Methodological Practices

- **Data splitting.** An independent held-out test set is mandatory. An 80/20 development/test split is recommended for independent samples. Use group-aware or time-aware splitting when appropriate.
- **Cross-validation.** Use shuffled 5-fold `KFold` for regression and `StratifiedKFold` for classification when appropriate, with `random_state=42`; use group/time-aware CV for dependent observations.
- **Pipelines.** Fit preprocessing and feature selection within training folds only.
- **Tuning.** Use `GridSearchCV` or a comparable cross-validation search; report parameter grids and primary selection metric.
- **Test data.** The held-out test set must not be used to choose algorithms, hyperparameters, features, preprocessing strategies, or classification thresholds. Evaluate the final selected configurations only after model development is complete.
- **Reproducibility.** Document random seeds, software versions, data access, and modeling settings.
- **Interpretation.** Do not make unsupported causal or clinical claims.

6 Deliverables

Deliverable 1: Jupyter Notebook(s). Submit executable notebook(s) containing EDA, preprocessing, training, cross-validation, tuning, evaluation, figures, and interpretations. Notebooks must run from beginning to end.

Deliverable 2: Scientific Report. Submit a PDF of approximately 5–7 pages, excluding references and appendices. Include: (1) Introduction, (2) Data and Methods, (3) Results, (4) Discussion, (5) Conclusions, and (6) References.

Deliverable 3: Reproducibility Information. Include a `README.md` with Python version, packages, data access instructions, execution steps, and any necessary files. For restricted data, describe an authorized reproducibility procedure rather than sharing protected data.

Deliverable 4: Presentation. Deliver a **5-minute presentation** with approximately 4–5 slides during the final exam time. Cover the biological question and data, methods and evaluation design, key results (including model complexity), and biological interpretation, limitations, and conclusions. Submit the slides with the other deliverables.

7 Grading Rubric

The project proposal is graded out of 10 points and requires instructor approval before the main analysis.

Component	Points
Project proposal	**10**
Biological background and research question	2
Dataset description and suitability	2
Selection and justification of algorithms	2
Preprocessing, validation, and evaluation plan	3
Feasibility within three weeks	1
Part 1. EDA and preprocessing	**15**
Dataset understanding and visualizations	6
Data splitting and preprocessing pipelines	6
Explanation of preprocessing and leakage	3
Part 2. Model development	**20**
Correct implementation of selected algorithms	8
Cross-validation and hyperparameter tuning	7
Justification of model choices	5
Part 3. Model evaluation and comparison	**20**
Appropriate metrics and experimental design	8
Cross-validation and independent test results	7
Figures and interpretation	5
Part 4. Model complexity and overfitting	**10**
First systematic experiment	4
Second systematic experiment	4
Interpretation of generalization behavior	2
Part 5. Model interpretation and biological discussion	**15**
Model comparison	4
Feature interpretation	5
Biological discussion, limitations, recommendations	6
Part 6. Scientific communication and reproducibility	**10**
Report organization and clarity	4
Reproducible, documented code	3
Oral presentation (5 minutes)	3
Total	**100**

8 Expectations and Evaluation Philosophy

This project emphasizes **correct methodology, critical thinking, and scientific interpretation**. You will not be graded primarily on the highest accuracy, ROC AUC, or R^2. An ambitious project with invalid evaluation is less successful than a focused project with rigorous methods. The proposal is a **10-point graded component** and also requires instructor approval before the main analysis.

Central question: How can appropriate machine learning methods address a meaningful biological question, and what can we learn by comparing their performance and behavior?